

SWAYAM KAMBLE

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🌐 [swayam-kamble](https://swayam-kamble.github.io)

📁 [Portfolio](#)

TECHNICAL SKILLS

Programming: Python, C, C++, Arduino, Embedded C

Technologies: IoT, Microcontrollers (ESP32, Arduino), Sensors & Actuators, MQTT, Wi-Fi, Bluetooth

AI/ML: Machine Learning, Deep Learning, LLMs, LangChain

Web Development: HTML, CSS, JavaScript

EXPERIENCE

5G Lab, VNIT Nagpur

2024 – Present

Research Intern

Nagpur, India

- Working on projects involving 5G and IoT under the guidance of Dr. Prabhat Sharma, focusing on **advanced wireless communication and intelligent connected systems**.

Signal Processing and Communication Research Center (SPCRC), IIIT Hyderabad

July 2024

Summer research Intern

Hyderabad, India

- Worked under Dr. Sachin Chaudhari on **IoT technologies and signal processing applications**.
- Developed an ultrasonic-based water level monitoring system with **temperature-compensated sensing** for accurate liquid level detection.

E-Yantra Lab, St. Vincent Pallotti College

2022 – Present

Workshop Conductor & Mentor

Nagpur, India

- Conducted workshops on IoT and emerging technologies for junior students, **sharing hands-on knowledge and real-world applications**.
- Mentored peers in core concepts of IoT, embedded systems, and AI integration, **fostering collaboration and technical growth**.

PROJECTS

e-Yantra Lab Automation System | ESP32, Python, Computer Vision, IoT Cloud

2024

- Developed an AI and IoT-based lab automation system using **computer vision for occupancy detection and smart device control**.
- Implemented facial recognition for attendance logging and secured access with real-time monitoring.
- Secured 1st place** among 520+ colleges in the eLSI Hackathon at IIT Bombay.

Smart Street Light Monitoring System | ESP32, ESP-NOW, Web Dashboard

2024

- Designed a multi-node architecture with **ESP32 controllers and ESP-NOW protocol** for low-latency communication.
- Created an intuitive web interface for real-time monitoring of streetlight operational status.
- College Startup Expo** with INR 10,000 prize for Smart Infrastructure Innovation.

Smart Home Environmental Monitoring System | ESP32, IoT Cloud, ML, Sensors

2024

- Built an advanced IoT solution integrating **multiple environmental sensors, cloud dashboards, and predictive analytics**.
- Implemented LSTM model for temperature/humidity forecasting with RMSE analysis for accuracy verification.
- Integrated with **ThingSpeak, oneM2M, MQTT, and Grafana** for comprehensive data visualization and control.

EDUCATION

St. Vincent Pallotti College of Engineering and Technology

2022 – Present

Bachelor of Technology (B.Tech) - Industrial Internet of Things (IIoT) — CGPA: 7.71/10 *Nagpur, India*

St. Paul Junior College

2020 – 2022

Higher Secondary Education (12th Grade) — Percentage: 75%

Nagpur, India

ACHIEVEMENTS

eLSI Hackathon, IIT Bombay

2024

1st Place Winner

National Level

- Led a team of 4 to develop an innovative AI and IoT-based lab automation system.
- Competed against 520+ colleges nationwide and secured first position.

College Startup Expo, Engineers Day

2024

Smart Infrastructure Innovation Award

College Level

- Received INR 10,000 cash prize for the Smart Street Light Monitoring System project.

CERTIFICATIONS

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| • Python Certification - CodeChef | • C++ Programming - DiSHA Computer Institute |
| • Web Development - Great Learning | • Machine Learning - Prutor IIT Kanpur |
| • Hands-on IoT - NPTEL | • Industrial Approach in Electronics |
| • NPTEL courses in Manufacturing Process Technology | • Sensors and Actuators - NPTEL |
| • Microprocessors and Microcontrollers | |